A Decision-focused Framework for Integrated Inventory Management under Heterogeneous Demand

1. Background and Motivation

In highly competitive retail environments, firms must manage thousands of distinct products while balancing service levels and operational costs. A central operational challenge is inventory control: determining when to reorder and how much to replenish. Larger order quantities and shorter review cycles reduce stockouts and lost sales but increase holding and ordering costs. Conversely, lean inventory policies reduce capital tied up in stock but expose the firm to service failures during demand spikes.

Traditional optimization methods, such as dynamic programming and stochastic programming, become computationally intractable, especially when integrating **interdependent constraints** across thousands of items. Moreover, these approaches often rely on rigid assumptions about demand distributions and replenishment structures, which rarely hold in practice. (Theodorou, 2023).

Furthermore, the sheer **heterogeneity of demand patterns**—ranging from smooth, high-volume staples to erratic, "lumpy" seasonal goods—precludes a one-size-fits-all optimization strategy. Managing thousands of items identically is not only inefficient but creates systemic risk. This necessitates a **strategic clustering approach** (Yılmaz, 2025) to categorize products based on their volatility and predictability. By segmenting the inventory, firms can differentiate their risk tolerance, applying lean principles to stable goods while prioritizing **system viability** and robustness for highly uncertain items.

To address these challenges, firms increasingly adopt Machine Learning (ML) models to forecast demand and then input predictions into Mathematical Programming (MP) solvers. However, the conventional “Predict-then-Optimize” framework suffers from a critical weakness: small forecast errors can lead to disproportionately large cost increases when the optimization model is sensitive to prediction inaccuracies. Minimizing statistical loss (e.g., MSE) does not necessarily improve economic decision quality.(Lahoud, 2025).

This motivates the shift toward a “Predict-and-Optimize” (PaO) paradigm (Lahoud, 2025), where predictive models are trained directly to minimize downstream operational costs rather than purely statistical forecast error.

2. Problem Definition

This project studies a multi-item inventory control problem under demand uncertainty, where a decision maker must determine replenishment quantities for thousands of products subject to:

- Asymmetric cost structure (holding, ordering, and lost-sales costs),
- Demand heterogeneity (including lumpy and erratic demand patterns),
- Global operational constraints (e.g., total budget and warehouse capacity).

The core research problem is:

> How can we design a decision-focused, data-driven optimization framework that directly minimizes total economic cost while balancing efficiency and robustness in a highly uncertain retail environment?

This problem is particularly interesting because it involves multiple fundamental trade-offs:

(1) Prediction Accuracy vs. Decision Quality

- **The Paradigm Flaw:** The conventional "Predict-then-Optimize" (PaO) paradigm completely decouples the machine learning forecasting model from the downstream inventory optimization task.
- **The Core Problem:** Traditional ML models aim to minimize the Mean Squared Error (MSE). However, in a real-world business context, the penalty for stockouts (c_p) and the cost of holding inventory (c_h) are highly asymmetric. A prediction model with a minimal MSE that systematically underestimates demand could lead to catastrophic stockout losses. Therefore, **low prediction error does not equate to high decision quality** (Lahoud, 2025). The challenge is how to make the neural network directly perceive "downstream decision costs" during training, rather than merely fitting the data distribution.

(2) Cost-Efficiency vs. Robustness

- **The Paradigm Flaw:** Even if a model calculates the cost-optimal order quantity based on expected values, the system remains fragile when facing items with extreme volatility (e.g., items with a Coefficient of Variation squared,$CV^2 > 0.49$).
- **The Core Problem:** Pursuing expected cost-efficiency blindfolds the decision-maker to extreme "black swan" demand spikes. Decision-makers must balance the trade-off between minimizing average expected costs and maintaining system robustness by establishing a "safety margin" (e.g., considering the 95th percentile worst-case scenarios) (Yılmaz, 2025).

(3) Local Optimization vs. Global Constraints

- **The Paradigm Flaw:** Solving the standard Newsvendor problem for a single SKU is mathematically trivial. However, simply aggregating local optimal solutions across thousands of items often violates physical real-world restrictions.
- **The Core Problem:** Local optimization at the individual item level inevitably clashes with global constraints, such as total warehouse capacity or procurement budgets. The engineering challenge lies in utilizing surrogate modeling to rapidly evaluate strategies across massive, heterogeneous datasets and finding a near-optimal joint strategy under global constraints (Theodorou, 2023).

3. Problem Description

3.1 Conceptual Model: The Integrated Inventory Intelligence System

The **Conceptual Model** represents a "Closed-loop Decision System" where prediction and optimization are no longer separate silos but a unified entity.

- **The Decision Maker (DM):** Acts as a "Strategic Replenishment Agent." It consists of a **Neural Network (Predictor)** that interprets complex market signals (price, promotions, seasonality) and a **Heuristic Solver (ABCA)** that translates those signals into actionable orders.
- **The Complicated Environment:** Unlike static models, this environment is a "Black-box" reflecting the **Walmart M5** reality. It is characterized by **stochastic demand**, where thousands of SKUs compete for a **limited global budget** and **finite shelf space**. Costs are non-linear and asymmetric, meaning the "pain" of a stockout is far greater than the "burden" of excess stock.
- **The Operations:** 1. **Risk-Aware Clustering:** Segmenting items into "Predictable" and "Volatile" groups to tailor the optimization strategy.
  2. **Adaptive Forecasting:** Generating demand estimates that "know" the cost of being wrong.
  3. **Heuristic Search:** Navigating the discrete search space of (s, S) levels to find a feasible, low-cost solution.
- **The Trade-offs:** The DM must constantly balance **Capital Liquidity** (don't buy too much) against **Service Level Integrity** (don't run out). It must also trade **Computational Precision** for **Speed**, utilizing a LightGBM Surrogate to bypass the "gradient-dead" nature of complex heuristic solvers.

3.2 Mathematical Model: Decision-Focused Optimization

In this model, we move beyond minimizing error ($\hat{D} - D$) to minimizing **Economic Regret**.

A. Decision Variables

Let i be the index for an SKU and t be the time period.

- $y_{it} \in R^{+}$: The predicted demand generated by the Decision Maker (Neural Network).
- $Q_{it} \in Z^{+}$: The order quantity determined by the Solver (ABCA) based on y_{it}.
- S_{it}: The target "Order-to-Level" (inventory ceiling) for item i.

B. The Objective Function (The "True" Loss)
The goal is to minimize the **Total Operational Cost (TC)**, which serves as our **Decision-focused Loss Function**:

$$min\, TC = \sum_i \sum_t (c_h * I_{it}^{+} + c_u * I_{it}^{-} + c_{fix} * \delta(Q_{it})$$

Where:

- $I_{it}^{+} = \max(0, Inventory\ on\ hand)$: Representing **Holding Costs** (c_h).
- $I_{it}^{-} = \max(0, Shortage)$: Representing **Underage/Stockout Costs** (c_u).
- $\delta(Q_{it})$: An indicator function that is 1 if $Q_{it} > 0$, representing **Fixed Ordering Costs**.

C. Operational Constraints

1. **Storage Constraint (The "Bag Limit"):**

   $$\sum_i v_i \cdot (I_{it-1} + Q_{it}) \leq v_{max}$$

   (Total volume of all items v_i must not exceed warehouse capacity v_{max}).
2. **Budget Constraint:**

   $$\sum_i p_i \cdot Q_{it} \leq B_{total}$$

   (Total procurement cost at price p_i must stay within the weekly budget B).
3. **Inventory Flow Balance:**
   $I_{it} = I_{it-1} + Q_{it} - D_{it}$
   (Ending inventory equals beginning inventory plus orders minus realized demand D).

D. The PaO Connection (The Gradient Link)
Because the mapping from $y_{it} \rightarrow Q_{it}$ is performed by a non-differentiable **Heuristic Solver (ABCA)**, we introduce the **Surrogate Gradient**:

$\nabla_{weight} \approx \frac{\partial TC}{\partial Q} \cdot \frac{\partial Q_{surrogate}}{\partial y} \cdot \frac{\partial y}{\partial weight}$ This allows the Neural Network to "see" through the complexity of the inventory environment and adjust its predictions to minimize TC.

4. Data Collection

4.1 Case Background: Walmart M5 Dataset

This study uses the Walmart M5 Forecasting dataset as the empirical foundation for the proposed inventory optimization framework. The dataset contains 1,913 days of daily unit sales observations for 3,049 products across 10 retail stores located in three U.S. states, namely California, Texas, and Wisconsin. Such a large-scale retail dataset is particularly suitable for this research because it captures substantial heterogeneity in demand behavior,

ranging from stable, high-volume household staples to highly volatile and intermittent seasonal products. This diversity is essential for evaluating whether a decision-focused inventory system can remain effective under both predictable and non-stationary demand conditions.

Beyond its scale, the Walmart M5 dataset is also valuable because it provides rich contextual information that can support both forecasting and operational decision-making. In addition to historical sales records, the data include price information, calendar effects, and event-related variables, making it possible to model demand as a function of both historical consumption and exogenous retail signals. This characteristic is especially important in the present study, as the objective is not limited to statistical prediction, but extends to the construction of a decision-oriented inventory environment in which demand forecasts directly affect replenishment outcomes.

4.2 Information Collection and Parameter Generation

Although the Walmart M5 dataset is originally designed for demand forecasting, it does not directly provide all the elements required for an inventory optimization problem. Therefore, this study transforms the raw sales data into a decision-oriented operations research environment by constructing demand, cost, capacity, and budget parameters that are compatible with replenishment analysis.

First, historical daily sales are aggregated into weekly demand profiles, denoted by D_{it}, where i indexes the stock-keeping unit (SKU) and t indexes the review period. Weekly aggregation is adopted because the proposed replenishment framework operates at the weekly decision level, which is more consistent with practical retail review cycles than day-level ordering. At the same time, exogenous demand drivers are extracted from the original dataset, including price changes, calendar effects, SNAP-related indicators, and promotional or event variables. These features are incorporated into the forecasting component of the framework so that the predictor can respond not only to historical sales patterns but also to external market signals that may alter short-term demand.

Second, the study constructs economic parameters to quantify the asymmetric costs associated with replenishment errors. The underage cost, denoted by c_u, is approximated by the unit profit margin, defined as the difference between the selling price and the estimated procurement cost. This parameter reflects the economic loss associated with stockouts and unmet demand. In contrast, the overage cost, denoted by co, is generated using a 20% annual holding-rate assumption combined with shelf-space opportunity costs. This parameter captures the financial burden of carrying excess inventory, including storage, capital occupation, and the opportunity cost of limited shelf capacity. In addition, a fixed ordering cost is incorporated whenever an order is placed, thereby reflecting the practical fact that replenishment decisions often incur setup or transaction costs independent of the exact order quantity.

Third, physical and financial constraints are generated to ensure that the replenishment problem remains grounded in realistic retail operating conditions. Product volume, denoted by v_i, is estimated based on product category, allowing broad classes of items such as household goods and hobby-related products to be assigned differentiated space

requirements. Warehouse capacity, denoted by Vmax, is set at the 80th percentile of historical total volume in order to simulate a congested storage environment and impose meaningful competition among SKUs for limited space. Meanwhile, the global budget constraint, denoted by B_{total}, is calculated as a rolling average of historical procurement costs. This design forces the replenishment solver to allocate limited capital strategically across multiple products rather than trivially increasing inventory for all items simultaneously.

Through these transformations, the original forecasting dataset is converted into an inventory optimization environment that includes four essential components: weekly demand signals, predictive features, economic cost parameters, and operational constraints. This enriched environment enables the subsequent methodology to evaluate decisions on the basis of actual business implications rather than forecast accuracy alone.

5. Method

5.1 Technical Stack

The proposed framework is implemented in Python and combines machine learning, surrogate modeling, and heuristic optimization within a unified decision-focused architecture. PyTorch is used to build the neural forecasting component because of its flexibility in defining custom loss functions and supporting gradient-based model updates. The surrogate module is trained using LightGBM within the Scikit-learn ecosystem in order to efficiently approximate the non-differentiable inventory environment. For benchmarking and small-scale validation, Gurobi is employed to solve tractable sub-problems exactly. However, because the large-scale replenishment problem involves thousands of heterogeneous SKUs under discrete and nonlinear constraints, the main optimization engine is a self-developed heuristic solver based on a customized Artificial Bee Colony Algorithm (ABCA).

5.2 Overall Solution Pipeline

The methodological contribution of this study lies in the development of an end-to-end Predict-and-Optimize framework that directly links demand forecasting with replenishment decision-making. Instead of treating prediction and optimization as two isolated stages, the proposed system forms a closed-loop decision architecture in which downstream inventory cost is fed back into the learning process of the predictor. This design allows the model to optimize decision quality rather than merely minimizing statistical forecast error.

The entire solution pipeline consists of four main stages: strategic clustering of SKUs, surrogate model training, heuristic replenishment optimization, and end-to-end feedback learning.

5.3 Strategic Clustering of Demand Patterns

Before forecasting and optimization, the study applies a risk-aware clustering procedure to segment products according to their demand behavior. Specifically, K-Means clustering is conducted on the basis of two widely used demand descriptors: the Average Demand

Interval (ADI) and the squared Coefficient of Variation (CV^2). Using these indicators, items are classified into four broad categories: Smooth, Erratic, Lumpy, and Constant.

This clustering stage serves both analytical and operational purposes. From an analytical perspective, it allows the study to explicitly recognize that retail demand is heterogeneous rather than homogeneous across all products. From an operational perspective, the segmentation enables the framework to assign differentiated risk sensitivities to distinct demand groups. For example, highly intermittent and volatile items may require more conservative replenishment logic or stronger penalty weights on underage outcomes, while smooth-demand items may allow more aggressive inventory compression. Therefore, clustering provides the foundation for a more adaptive and robust optimization framework, especially under non-stationary demand conditions.

5.4 Surrogate Cost Modeling

A central methodological challenge in this research is that the inventory environment is not differentiable. The mapping from predicted demand to replenishment action is mediated by a heuristic solver, and the resulting cost function is shaped by discrete ordering rules, nonlinear constraints, and asymmetric penalties. As a result, the true decision pipeline cannot be optimized directly through conventional gradient descent.

To address this difficulty, the study introduces a surrogate cost evaluator based on LightGBM. The surrogate model is trained on a large set of simulated inventory scenarios, each of which contains a combination of demand predictions, replenishment decisions, and realized cost outcomes. In effect, the surrogate learns an approximate mapping of the form $(\text{Prediction}, \text{Decision}) \rightarrow \text{Cost}$. Once trained, it functions as a computationally efficient proxy for the otherwise opaque inventory environment.

The role of the surrogate is not to replace the replenishment solver itself, but to provide a learnable approximation of the cost response surface. In this sense, it acts as a “digital twin” of the operational environment, enabling the neural network to receive structured feedback on the economic consequences of its forecasts. This mechanism creates a practical bridge between predictive modeling and decision-focused learning in settings where exact gradients are unavailable.

5.5 Self-Developed Heuristic Solver: Customized ABCA

The replenishment optimization problem is solved by a customized Artificial Bee Colony Algorithm. This solver is designed to handle the high dimensionality and discrete nature of the weekly replenishment problem, where order decisions must satisfy budget and storage constraints across thousands of SKUs simultaneously. Compared with standard linear solvers, the ABCA is better suited to the present setting because it can explore complex (s,S)-type inventory policies under nonlinear, combinatorial conditions.

The algorithm contains three complementary search roles. First, employed bees explore the local neighborhood of current (s,S) inventory levels in order to identify cost-reducing adjustments. Second, onlooker bees allocate additional search effort toward high-performing SKU combinations that remain feasible under the global budget and warehouse capacity

constraints. Third, scout bees reset underperforming decision vectors, thereby improving exploration and reducing the risk of premature convergence to local optima. Through the interaction of these three roles, the ABCA performs adaptive search over a large, constrained decision space and produces replenishment quantities Qit for each review period.

5.6 End-to-End Decision-Focused Feedback Loop

The final stage of the methodology is the end-to-end feedback loop that connects forecast generation to operational evaluation. After the neural network outputs predicted demand, the ABCA solver determines the replenishment decision vector. The quality of this decision is then evaluated by comparing the solver's output against realized demand and computing the corresponding total operational cost, including underage, overage, and ordering costs. This decision-focused cost serves as the true training signal of the system.

Because the true solver is not differentiable, the gradient needed for neural network training is approximated through the surrogate model. In other words, the surrogate enables the cost signal to be propagated backward from the replenishment outcome to the forecasting weights. As a result, the predictor is trained to produce forecasts that are economically useful rather than merely numerically accurate. This is a key distinction from traditional Predict-then-Optimize pipelines, in which the forecasting model is usually trained with mean squared error and remains blind to the downstream decision consequences of its errors.

Formally, the framework follows a decision-focused objective in which the learning target is the operational cost induced by the replenishment decision:

$$Loss = \text{Cost}(Q^*(\hat{D}), D_{true}) .$$

Under this formulation, the model learns to avoid economically damaging prediction errors, especially those that create costly stockouts for high-margin or highly volatile items. In this way, the proposed methodology aligns predictive learning with managerial objectives and provides a more realistic basis for retail inventory decision-making.

6. Expected Outcomes

This research is expected to achieve the following core outcomes:

6.1 Methodological Innovations

Develop an integrated "Predict+Optimize" inventory decision framework that enables neural networks to directly learn to minimize inventory costs, rather than traditional prediction errors.
Design a lightweight surrogate model to solve the technical challenge of non-differentiability in heuristic algorithms, enabling end-to-end training of the entire pipeline.
Establish a product categorization mechanism based on demand volatility to match differentiated inventory strategies for different product types.

6.2 Empirical Results

Validation on the Walmart M5 dataset (3,000+ products) demonstrating, compared to traditional "Predict-then-Optimize" approaches:
Significant reduction in total inventory costs (holding costs + stockout costs + ordering costs). Specifically, the combined effect of clustering data and PaO is expected to be approximately 15%-25% higher than using only the baseline predict-then-optimize optimization.
More robust system performance during periods of drastic demand fluctuations such as promotional seasons.

6.3 Deliverables

Open-source Python code library: A complete implementation encompassing data processing, decision-focused neural networks, heuristic solvers, and surrogate models.
A practical guide for retail enterprises: Insights on calibrating cost parameters and setting safety stock levels based on product characteristics.

6.4 Practical Application Value

Assist retailers in achieving intelligent replenishment for thousands of products under limited budget and warehouse space constraints.
Prevent stockouts for high-margin products while reducing the accumulation of slow-moving items, balancing service levels and capital efficiency.
Provide explainable decision rationales, enabling managers to understand "why this product requires additional inventory."

References

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